INTERNATIONAL news

Tremendous growth expected in global LED market

The LED market is expected to grow by $25.78 billion during 2020-24. The LED market is driven by the declining manufacturing cost of LEDs. LED lights is now one of the fastest growing businesses in the world, and India is considered the second biggest consumer of these lights. It is estimated that around 50 million LED bulbs are produced in India every month in the organised sector, and production of around 20 million LED lamps takes place in the unorganised sector.

GaN semiconductor device market growing substantially

The global GaN semiconductor device market will grow to $24.9 billion by 2026 from $19.4 billion in 2021 at a CAGR of 5.2 percent between 2021 and 2026. The key drivers fueling the growth of this market include the wide gap property of GaN material facilitating innovative applications, success of GaN in RF power electronics, and the increasing adoption of GaN RF semiconductor devices for defence and aerospace applications. Asia-Pacific region is expected to hold the largest share of the GaN semiconductor device market.

Investments in China's fab capacity expansion to exceed $160bn

China has made some progress in developing its domestic fabless industry, though it is still dependent on importing the final manufactured product in several cases. Investments in new fabs or capacity expansion have been forecast to exceed $160 billion in China over the coming five to seven years. Geopolitical developments in recent years, particularly the inclusion of several Chinese entities (notably Huawei, Hikvision amongst others) by the US on its Restricted Entity list may have added urgency to China's initiatives to localise the tech industry and reduce its import dependence.

Production of PCs could be hit due to chip shortage

In the wake of the chip supply crisis which has doomed production plans of various manufacturers, Dell Technologies Inc and HP Inc have said that PC supplies could be the next victim of the ongoing chip crunch this year. The companies revealed that the shortage could impact their ability to meet demand for laptops. Global shipments of PCs grew 55.2 percent during the first quarter.

Foxconn goes for EV partnership with Gigasolar

Foxconn said a subsidiary has invested T$995.2 million (US$36 million) in Gigasolar Materials Corp to develop electric vehicle (EV) battery materials. The investment via a private placement through a Taiwan based subsidiary will make Foxconn the second-largest shareholder in Gigasolar, known for manufacturing solar cell materials. The Taiwanese company aims to provide components or services to ten percent of the world's electric cars by 2025.

Volkswagen looking for battery material suppliers for EVs

German automotive manufacturer Volkswagen is accelerating its entry in the expanding electric mobility market as it is in talks with battery materials suppliers to secure direct access to raw materials for electric vehicle batteries via partnerships, board member Thomas Schmall revealed. Volkswagen is trying to exert more control over key components in its supply chain such as semiconductors and lithium so that it can overcome any potential bottlenecks and keep its plants running at full capacity.

TSMC planning to build a semiconductor fab in Japan

TSMC is said to be formulating plans of building a semiconductor foundry in Japan. The Taiwan based semiconductor giant might move ahead with the plan if the conditions between the United States and China continue to deteriorate. The new plant, as per the experts, is expected to use 30.4cm (12-inch) wafers and produce 16nm and 28nm products. It might fabricate system semiconductors used in automobiles and smartphones whose shortage is deepening in the world.

Google develops AI model to design chips much faster

A team of Google researchers is designing next-generation artificial intelligence (AI) chips. The team has created an AI model that allows chip design to be performed by artificial agents with more experience than any human designer. The model could generate a design that optimises the placement of different components on the chip in about six hours. Despite five decades of research, chip floor planning has defied automation, requiring months of intense effort by physical design engineers to produce manufacturable layouts.

MKS Instruments approaches Atotech with acquisition offer

MKS Instruments, a semiconductor equipment maker, has approached Atotech Limited, a specialty chemicals company, with an acquisition offer. The potential combination would expand MKS's offerings in chip manufacturing through the addition of Atotech's plating chemicals. Atotech has a market value of around $4.7 billion as of today. The company was floated on an IPO just a few months earlier by Carlyle Group Inc. However, it is not sure whether Atotech will entertain the offer made by MKS Instruments or not.

Samsung to supply foldable displays to Google and others

Smartphone makers Oppo, Xiaomi, and Google will be supplied with folding displays by Samsung for their purported foldable smartphone models. Samsung will start the mass production for its OLED panels from October this year in order to supply to these brands. Google, Oppo, and Xiaomi are planning to unveil their foldable smartphones in the fourth quarter of 2021. These panels are being made on the A3 lines in Samsung's Asan plant.

Pimoroni switches to larger Sheffield base for expansion

Electronics manufacturer Pimoroni has doubled its production, research and development, and warehousing space with a relocation move in Sheffield. Pimoroni Ltd, which was founded in 2012, has grown into one of the well-known manufacturers and suppliers of educational, industrial, and hobbyist electronics in the UK with over 100 distributors worldwide. The company, which offers 2,500 products, has a £6m turnover and 40 staff members, has brought all its operations under one roof with a 10-year lease at a 26,000sqft headquarter on Parkway Drive in Sheffield.

GlobalFoundries and GlobalWafers announce partnership

GlobalFoundries (GF) and GlobalWafers (GWC) have announced an $800 million agreement to add 300mm silicon-on-insulator (SoI) wafer manufacturing and expand existing 200mm SoI wafer production at GWC's MEMC facility in O'Fallon, Missouri, USA. The announcement comes at a time when the United States is seeking to fortify and expand its semiconductor supply chain. It is to be noted here that only twelve percent of the world's semiconductor manufacturing capacity is in the United States.

LG smartphone production lines to now make home appliances

Following its exit from the smartphone business, LG Electronics is now converting its overseas smartphone manufacturing factories into home appliances producing facilities. The $6 million expansion plan will provide the South Korean electronics giant with new production lines for laptops and monitors at the Manaus plant. The tech company has also converted its smartphone line in a plant in Vietnam to make home appliances.